

Ahuva Fogel

Software Engineer Holon, Israel | 054-854-9065 | ahuva9065@gmail.com
[GitHub Profile](#) | [\[LinkedIn Profile\]](#)

Professional Profile

Analytical and results-driven Software Engineer with a strong foundation in Object-Oriented Programming (OOP) and low-level software architecture. Proven track record of designing efficient algorithms, writing clean, maintainable code, and mastering complex technological systems through self-directed learning. Adept at collaborative teamwork in multidisciplinary environments and utilizing advanced tools to accelerate development cycles. Specialized in C++, Python, and systems-level engineering concepts.

Technical Skills

- **Programming Languages:** C++, Python, C#, Java, JavaScript, TypeScript, Assembly.
- **Operating Systems:** Linux, Windows.
- **Core Competencies:** Real-Time & Embedded concepts, Object-Oriented Design (OOD/OOP), Data Structures, Algorithms, Operating Systems, Computer Architecture, Digital Systems.
- **Technologies & Tools:** Git, Postman, Visual Studio, VS Code, PyCharm, IntelliJ IDEA.
- **Databases:** SQL Server, MySQL, PostgreSQL.

Education

MAHAT Engineering Diploma (Software Engineering) & Advanced Academic Coursework

Graduated | Combined Certification with **Lev Institute (JCT)**

- **Key Coursework Completed:** Advanced Programming & Object-Oriented Programming (OOP) in C++, Operating Systems, Computer Architecture, Digital Systems, Data Structures & Complexity Analysis, Introduction to Information Security, Basic Physics, and Electromagnetism.

Teaching Certification | Ministry of Education

- Certified Senior Instructor.

Selected Projects

Network Communication Simulator (C++ / OOP)

- Designed and implemented a high-performance network communication simulator modeling component-to-component message routing.
- Integrated advanced Object-Oriented Programming (OOP) principles to develop a highly modular, decoupled architecture.
- Programmed robust algorithms for real-time logic constraints, including queue management, packet transmission delays, and data-loss mitigation.

Strategic Board Game (JavaScript / Algorithms)

- Developed a complex strategic board game requiring robust move-validation algorithms and state synchronization.
- Implemented complex structural logic to govern real-time state mutations and game flow control.

Bakery Order Management System (Full-Stack / Databases)

- Engineered automated backend server logic and relational database schemas to safely process business transactions.
- Built secure data fetching structures and live dashboards, demonstrating handling of parallel processes and multi-layered system components.

Languages

- **Hebrew:** Native speaker.
- **English:** High professional proficiency.